HMC ONE
AI STRATEGY. BUILT TO SCALE.

CLIENT CASE STUDY · WIRES & CABLES MANUFACTURING · QUALITY

INSULATION LINES · HOUSE WIRE AND POWER CABLE · ROOT CAUSE & 8D

Root Cause Assistant

The same pinholes, thin insulation and off-centre conductors kept coming back. A root cause assistant now joins rejection logs with line data, ranks defects by rupees, scores every cause against evidence and drafts the 8D.

~~94.1~~ → **97.8** % first-pass yield on insulation lines



97.8 %
First-pass yield on insulation lines
▲ from 94.1%



62 % fewer
Spark faults per 100 km
▼ moderate and speed controls



38 % lower
Scrap cost on insulation lines
▼ fewer cut-outs, rejected drums

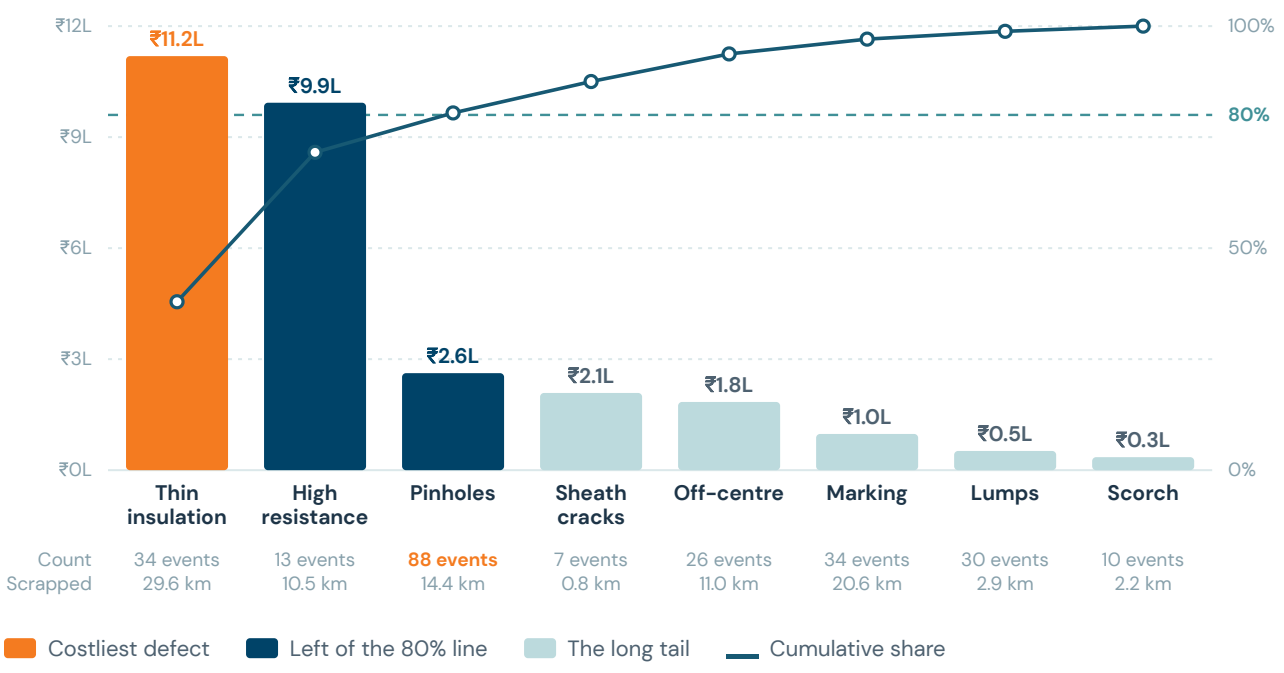


3 days
To close a root cause analysis
▼ from about 21 days

Rings: yield now with the earlier 94.1% inside; share of spark faults removed; share of scrap cost removed; 3 days against the earlier ~21 (inner ring). Figures indicative; validated metrics available on request.

01 The loudest defect is not the costliest

Auto Pareto by ₹ scrap · synthetic 30-day rejection log, 4 lines, as shipped with the tool



FIX FIRST
Thin insulation: ₹11.2L of ₹29.5L
Only 34 events, but each can scrap a whole coil: 29.6 km cut out in the month.

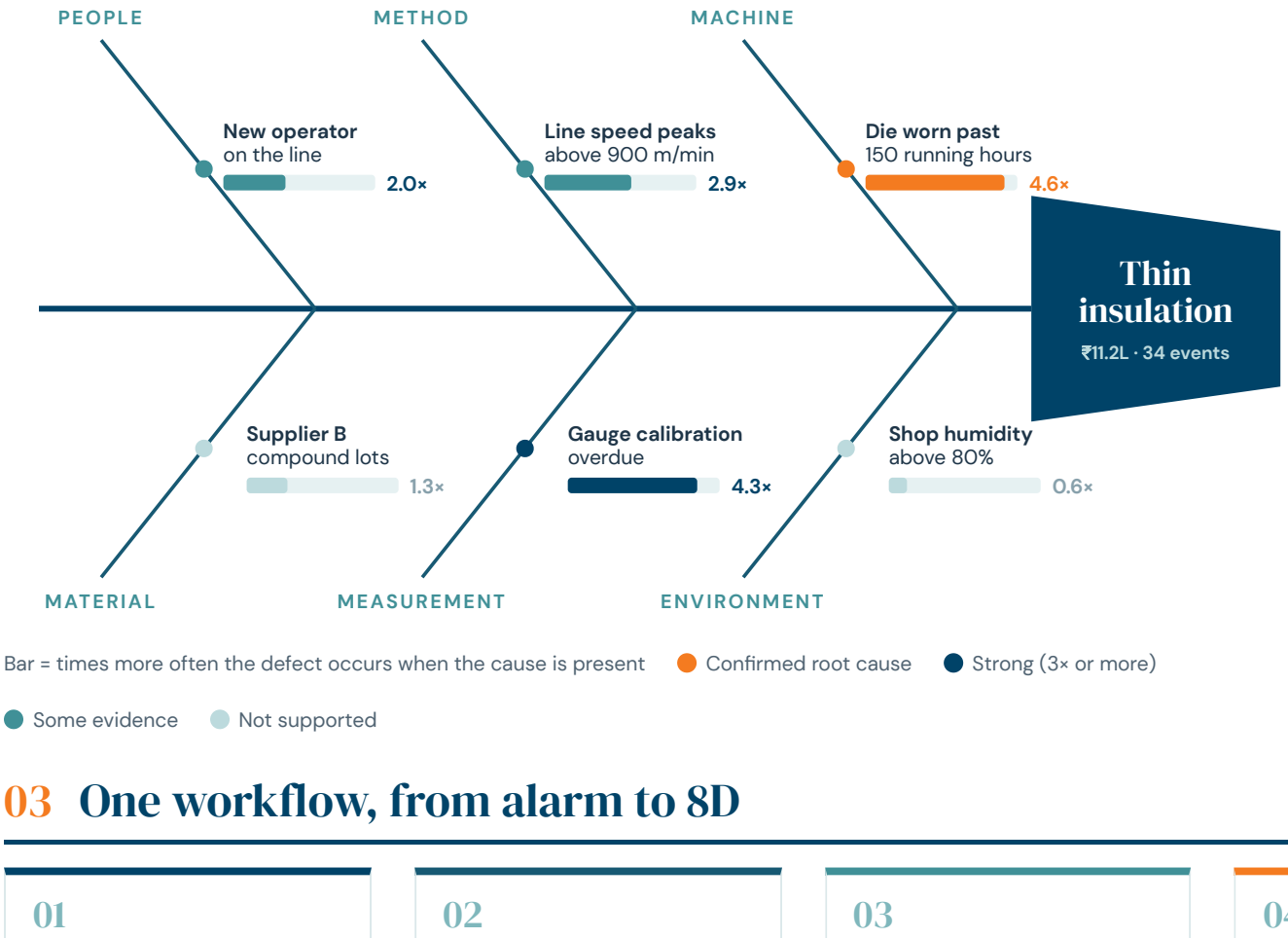
THE NOISE
Pinholes are seen most, cost third
88 spark faults, the most of any defect, yet ₹2.6L. The team used to chase these first.

CONCENTRATION
Two defects, 72% of scrap value
Thin insulation and high conductor resistance; the first three defects reach 80%.

RANK BY
Rupees, metres or count
Same log, three views. Money decides which defect gets an owner and an 8D this week.

02 Every fishbone cause, tested against the data

Evidence-scored fishbone for thin insulation · 360 synthetic production runs



5 WHYS · DIE WORN PAST 150 H
From the evidence to the root cause

1. Outer diameter drifted below spec as the die wore.
2. Dies are changed by calendar, not by running hours.
3. Die hours are not tracked on the line.
4. The PLC counter resets on every product changeover.
- R **No die-life limit in the control plan, so wear past 150 hours goes unnoticed until the gauge alarms.**

Contain: measure thickness on the last five drums per line; hold any below 0.62 mm.
Correct: change dies at 150 running hours.
Prevent: die-hour counter on the HMI; die life in the control plan.

03 One workflow, from alarm to 8D

The root cause assistant, module by module

01 Live line monitor
Diameter, eccentricity and spark faults per run; outer diameter spec 3.05–3.35 mm.
SEES THE FAULT

→

02 Auto Pareto
Rejections ranked by rupees, metres or count; the bars left of 80% get the attention.
PICKS THE PROBLEM

→

03 Evidence fishbone
6M cause library per defect, each cause written as a test on line data, with 5 Whys.
FINDS THE CAUSE

→

04 8D report
GenAI drafts the eight disciplines from the data and shop-floor notes; the engineer signs.
CLOSES THE LOOP

→

ROOT CAUSE ANALYSIS
~~21~~ → **3**
days to close. The 8D is drafted the same day.

04 Four pain points, four fixes

Before → after

BEFORE
Data in silos
Rejection registers, spark-tester counts and extruder settings in separate sheets and logbooks.

PYTHON JOIN

AFTER
One record per run
Rejections joined with line data: speed, zone temperatures, compound lot, die hours.

BEFORE
Fixing the loudest
Teams chased whatever failed last, not the defect costing the most metres and rupees.

AUTO PARETO

AFTER
Ranked by rupees
The Pareto opens every morning huddle; the top defect gets an owner and an 8D.

BEFORE
Fishbones from memory
Causes brainstormed in meetings; the root cause was the senior person's opinion.

EVIDENCE SCORE

AFTER
Every cause scored
Data, then 5 Whys, then verified on the control chart.

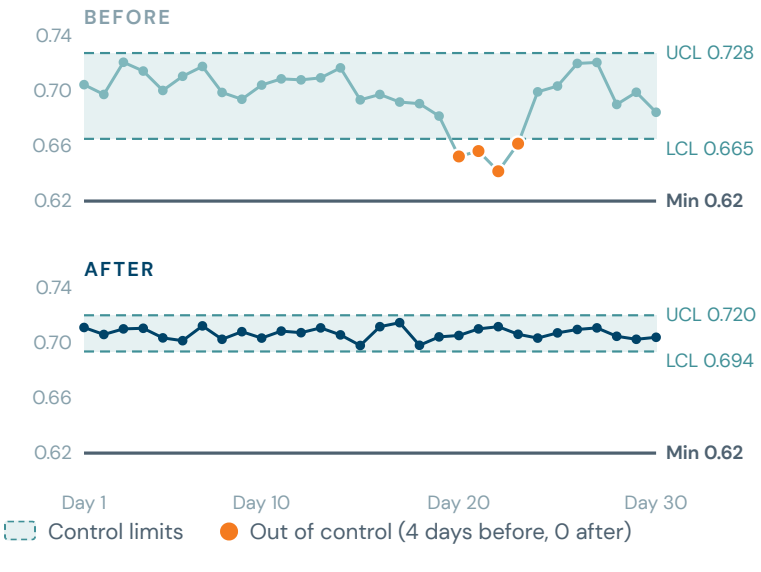
BEFORE
8D reports took weeks
Customer complaints waited about three weeks; repeat defects came back.

GENAI 8D

AFTER
Drafted the same day
Closed in about three days; control plan and PFMEA updated on closure.

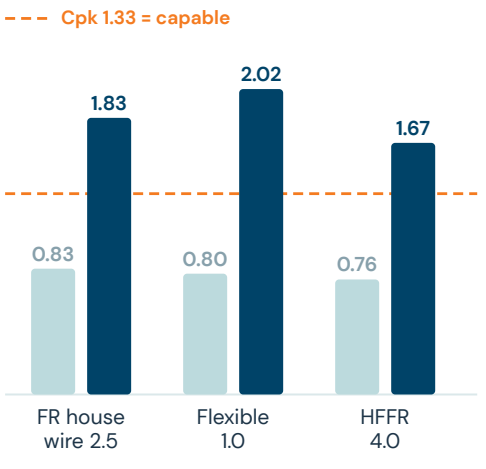
05 A process that stays in control

Insulation thickness, daily average (mm) · X-bar chart · indicative



06 Capability, Cpk

Before (light) and after (navy) · indicative



07 How HMC ran it

2 weeks — Discover
Six months of rejections walked through with QA; defect codes cleaned; scrap valued in metres and rupees.

2 weeks — Design
Every fishbone cause written as a test; 6M cause library, evidence score, 8D template agreed.

5 weeks — Build
Line view, auto Pareto, evidence-scored fishbone and GenAI 8D in one workflow.

2 lines — Adopt
Daily quality huddle; fixes verified on the control chart, not by feel.

All plants — Scale
Sheathing, cabling and wire drawing added; lot-level feedback to suppliers.

What any plant can borrow

01 Rank by rupees, not by noise
The Pareto in money terms tells the team which single defect to fix first.

02 Make every cause testable
A fishbone cause that can't be checked against data is an opinion. Write each one as a test.

03 Close the loop on the control chart
A root cause is confirmed only when the chart shows the process staying in control afterwards.

Built with **Python** · **Line PLC & gauge data** · **OpenAI GPT** · **D3** · **Power BI** · **SharePoint** | Prepared by Havish M Consulting · Figures indicative; validated metrics available on request.

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